

Car Details V4

Automotive · 2,059 records · 20 features · 18 Jul 2026

Records	Features	Completeness	Missing Values
2,059	20	98.2%	745

01 Executive Summary

This report presents a comprehensive analysis of the **Automotive** dataset comprising **2,059 records** across **20 features**. The dataset contains **8 numeric** and **12 categorical** columns, with **745 missing values** addressed during the cleaning phase. Key patterns, correlations, and actionable insights are documented below.

02 Objectives

- Explore the structure and quality of the Automotive dataset
- Identify distributions and statistical properties of key features
- Detect correlations, outliers, and anomalies in the data
- Generate actionable recommendations for data-driven decision making

03 Methodology

Data was ingested in CSV/Excel format and validated for structure integrity. Automated cleaning was applied to handle 745 missing values and 0 duplicate records. Exploratory analysis, statistical summaries, and correlation analysis were performed across all 20 features.

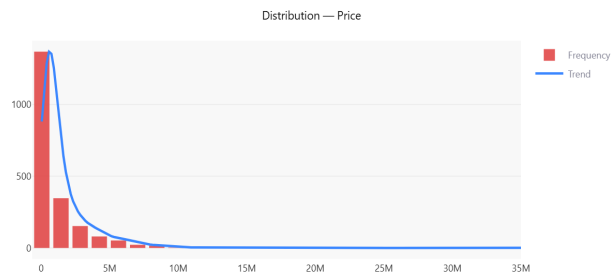
04 Statistical Snapshot

index	count	mean	std	min	25%	50%	75%	max
Price	2059.0	1702991.696	2419880.635	49000.0	484999.0	825000.0	1925000.0	35000000.0
Kilometer	2059.0	54224.714	57361.721	0.0	29000.0	50000.0	72000.0	2000000.0
Fuel Tank Capacity	1946.0	52.002	15.11	15.0	41.25	50.0	60.0	105.0
Seating Capacity	1995.0	5.306	0.822	2.0	5.0	5.0	5.0	8.0

05 Data Visualizations

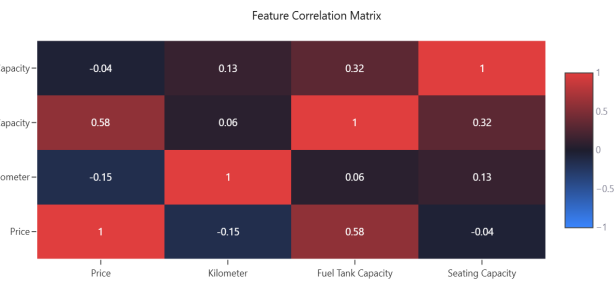
Charts are generated on the most statistically significant features of the dataset.

Distribution Analysis



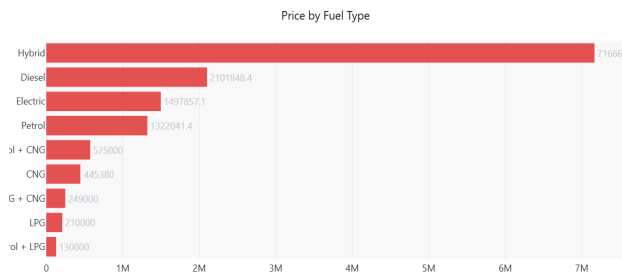
The distribution of **Price** shows a right skew (skewness = 4.97).
Mean: **1702991.70**, Median: **825000.00**.

Correlation Matrix



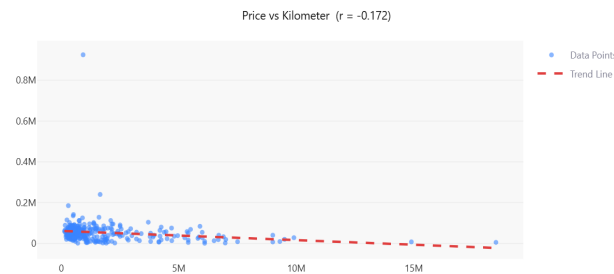
Strongest relationship: **Price** ↔ **Fuel Tank Capacity** ($r=0.58$).
High correlation indicates potential multicollinearity to address before modeling.

Category Breakdown



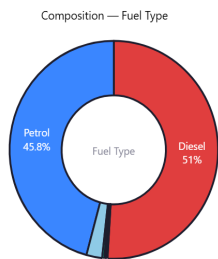
Hybrid leads with the highest average Price. Category analysis reveals significant variation across 9 groups.

Relationship Analysis



Price and **Kilometer** have a weak negative relationship ($r = -0.172$). The trend line shows the direction of linear dependency.

Composition Analysis



Diesel is the dominant category (51.0%) in **Fuel Type**. 9 distinct categories present.

06 Key Findings

- **Price**: Mean = 1702991.70, Std = 2419880.64, skewness = 4.97
- **Kilometer**: Mean = 54224.71, Std = 57361.72, skewness = 20.98
- **Length** and **Fuel Tank Capacity** show the strongest correlation ($r = 0.813$)
- Dataset completeness: **98.2%** of cells are populated

07 Conclusions

The Automotive dataset provides a solid foundation for data-driven analysis and modeling. After cleaning and exploration, the data reveals meaningful patterns across key features that can inform strategic decisions and future predictive modeling efforts.

08 Actionable Recommendations

1	Address remaining 745 missing values before building predictive models
2	Focus feature engineering on the top correlated columns identified in this report
3	Apply dimensionality reduction (PCA) if building ML models with all 20 features
4	Schedule periodic data refresh to maintain dataset timeliness and relevance

09 Suggested Next Steps

Clean & Enrich	Handle missing values and engineer new features
Build ML Model	Train on top features: Price, Kilometer, Fuel Tank Capacity
Monitor KPIs	Set up automated dashboards for key metrics
Share Insights	Distribute report to stakeholders for decisions